TATA MOTORS LIMITED

Company Research & Case Study

DIPLOMA FINAL-YEAR INTERNSHIP
COMPANY RESEARCH REPORT

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1. Introduction

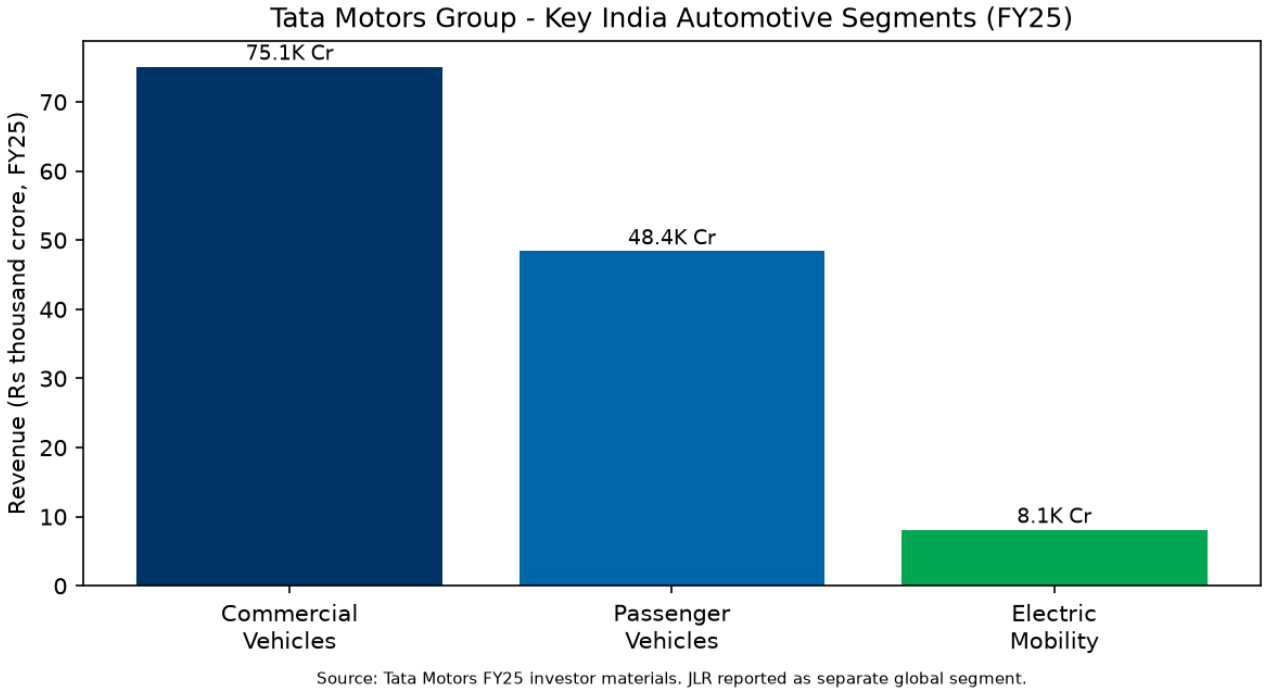
The automobile industry is vital to India's economy. Tata Motors is a major manufacturer in passenger vehicles, commercial vehicles, and electric mobility. This report studies its history, segments, products, manufacturing methods, R&D, and challenges - relevant to electronics and communication engineering students observing modern vehicle electronics.

2. About Tata Motors

Founded in 1945 as TELCO, Tata Motors designs, manufactures, and services cars, SUVs, trucks, buses, and EVs for consumers and fleets. It is part of the Tata Group and invests in safety, sustainability, and connected technology.

3. Business Segments

FY25 group segments include Tata Commercial Vehicles, Tata Passenger Vehicles, electric mobility, and Jaguar Land Rover. From October 2025, CV and PV businesses were demerged into focused listed companies per NCLT-approved scheme.



4. History

Key milestones: 1945 foundation; 1950s CV entry; 1998 Indica; 2004 Daewoo CV; 2008 JLR; 2010s–2020s SUV and EV growth; 2025 demerger.

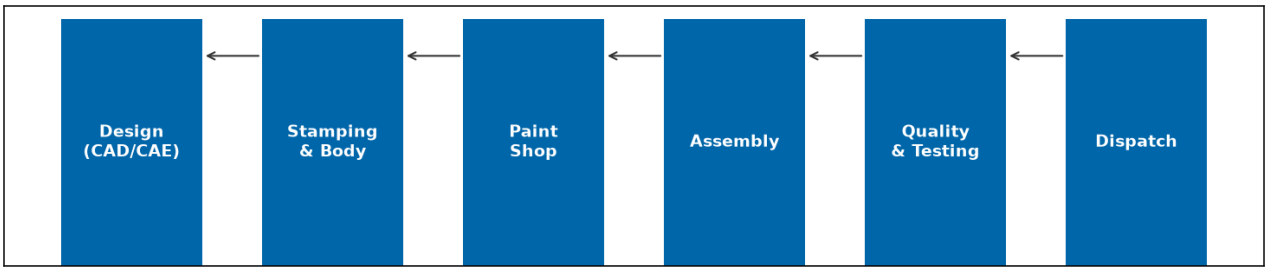
Milestones in Tata Motors History



5. Manufacturing

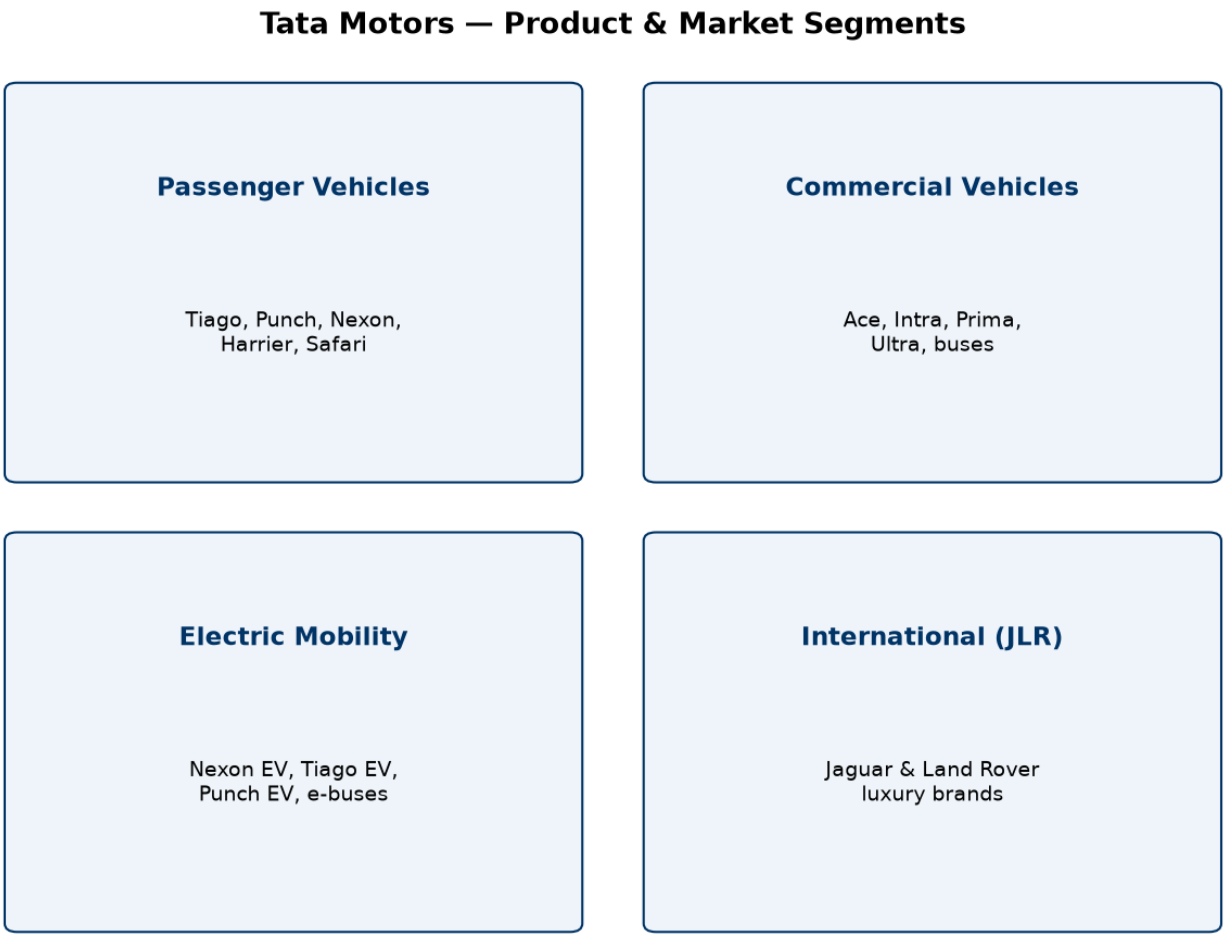
Lean, JIT, Kaizen, TQM, automation, CAD/CAM, supply chain management, environmental compliance, EV high-voltage safety, and end-of-line testing are central practices.

Typical Automobile Manufacturing Flow (Tata Motors & Industry Practice)



6. Products

Passenger: Tiago, Altroz, Punch, Nexon, Harrier, Safari. Commercial: Ace, Intra, Ultra, Prima, buses. Electric: Nexon EV, Tiago EV, Punch EV, Ace EV, e-buses.



7. Electronics & R&D

ECUs, BMS, infotainment, telematics, ADAS sensors, CAN/LIN/Ethernet networks, and OTA software are key ECE-related areas in modern Tata vehicles.

8–11. Plants, Challenges, Methodology, Conclusion

Major Indian plants include Pune, Jamshedpur, Lucknow, Sanand, and Dharwad. Challenges include competition, supply chain, electrification, and regulation. Research uses public annual reports and authorised internship observations. Tata Motors exemplifies scaled manufacturing with innovation in EV and safety.

References

[1] Tata Motors Integrated Annual Report 2024–25.
[2] Tata Motors Q4 FY25 investor materials.
[3] www.tatamotors.com